Which condensed structure(s) contain an ether functional group?

I. $CH_3CH_2CO_2CH_2CH_3$

II. $CH_3CH_2OCH_2CH_3$

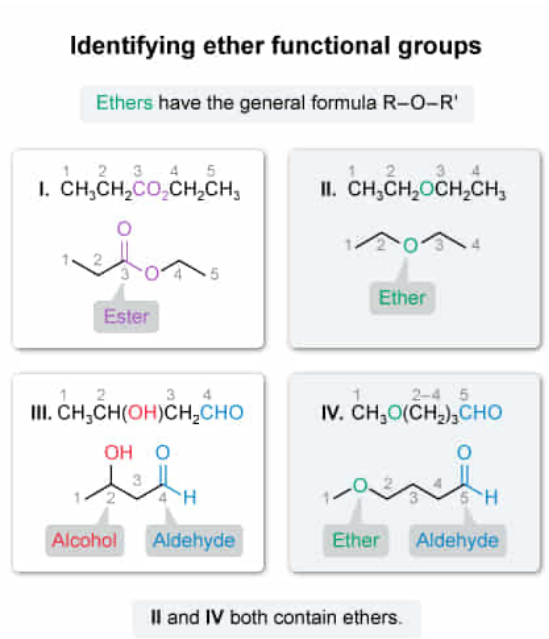
III. $CH_3CH(OH)CH_2CHO$

IV. $CH_3O(CH_2)_3CHO$

- A. I only [6%]
- B. II only [16%]
- ✔ C. II and IV only [69%]
- D. III and IV only [7%]

Correct
69% answered correctly

Explanation:



Although organic molecules are composed of primarily carbon and hydrogen atoms, they can contain other heteroatoms, including oxygen, nitrogen, and halogens. Functional groups represent recurring bonding patterns between atoms and are the reactive portions of a molecule used for **classification**. Ethers are an oxygen-containing group with the general formula **R–O–R'**, where the R groups are alkyl groups (the alkyl groups can be identical or different).

Organic molecules can be depicted in several ways. Condensed structural formulas show all atoms present in a molecule but do not explicitly show every bond. Branch points are

indicated by parentheses, and nonbonding electrons and multiple bonds are inferred based on the octet rule.

The condensed structure $CH_3CH_2OCH_2CH_3$ contains two ethyl groups (CH_3CH_2–) bonded to an oxygen atom **(Number II)**. $CH_3O(CH_2)_3CHO$ also has two alkyl groups, a methyl group (CH_3–) and a –$(CH_2)_3$– group, bonded to an oxygen atom **(Number IV)**. Therefore, **$CH_3CH_2OCH_2CH_3$** and **$CH_3O(CH_2)_3CHO$** both **contain an ether functional group**.

(Number I) The condensed structure $CH_3CH_2COOCH_2CH_3$ has one oxygen-containing functional group. Carbon 3 is bonded to carbon 2 and zero hydrogen atoms, indicating carbon 3 needs three more bonds to complete its octet. Therefore, carbon 3 must be bonded to both oxygen atoms listed after it in the condensed structure, and a double bond to one oxygen and a single bond to the other completes the octet of carbon 3. This bonding pattern is an ester functional group.

(Number III) The condensed structure $CH_3CH(OH)CH_2CHO$ has two oxygen-containing functional groups: the OH listed after carbon 2, which is a hydroxyl substituent (secondary alcohol, in this case), and CHO (carbon 5). Carbon 5 is bonded to carbon 4 and one hydrogen atom, indicating it needs two more bonds to complete its octet. A double bond to the oxygen atom listed after carbon 5 in the condensed structure completes its octet. Therefore carbon 5 is a carbonyl bonded to a H and an alkyl group, which is indicative of an aldehyde.

Educational objective:
Functional groups are common bonding patterns of atoms and are used to classify molecules. Ethers are an oxygen-containing functional group with the general formula R–O–R', where the R groups are two alkyl groups.

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Subject:	**System:**
Organic Chemistry	Introduction to Organic Chemistry